

Battered Fish — The Science-First Version

Shatter-crisp, lacy, never greasy — fish and chips by physics. Cold carbonated batter, rice flour for the shatter, oil hot enough to set it on contact.

Yield: 10 portions Active: 30 min Total: 40 min Oil: 185°C (365°F) Done: deep gold, crisp

WHERE IT WAS BORN Sephardic Jewish fried fish (<i>pescado frito</i>) met the fried chip in industrial Britain; by the 1860s the “chippy” was born and fish and chips became the national working meal.	EATEN, AUTHENTICALLY With thrice-cooked chips, mushy peas, malt vinegar and tartare — wrapped and eaten hot, the batter the star and the fish kept plain.	HOW IT’S USED NOW The chippy classic, plus tacos, a burger, a butty; the same cold batter fries onion rings, sausages and veg.
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INGREDIENTS

10 PORTIONS · WEIGHT

FISH

White fish fillets (<i>cod/haddock, skinless</i>)	3 lb 4.9 oz
Plain flour (<i>for dredging</i>)	$\frac{5}{8}$ cup

BATTER (MIX AT THE LAST MINUTE, COLD)

Plain flour	1 $\frac{1}{4}$ cups
Rice flour	1 cup
Baking powder	2 tsp
Fine sea salt	1 tsp
Cold lager / sparkling water (<i>ice-cold</i>)	1 $\frac{5}{8}$ cups
Vodka (<i>cold, optional</i>)	2 $\frac{3}{4}$ tbsps

THE RATIO
Flour 100% (half rice, half plain) · cold beer ~130% · baking powder 3% · salt 2%. A thin coating batter — but the *cold* and the *fizz* matter as much as the numbers.

Kit: deep pot / fryer · probe or fry thermometer · spider / slotted spoon · rack to drain (not a plate) · whisk

METHOD

- Prep the fish.** Pat very dry, cut into even 150g portions, salt lightly, keep cold.

Surface water repels batter and drops the oil temperature; even pieces fry evenly. — McGee 2004
- Heat the oil to 185°C (365°F).**

Hot enough to flash-set the crust and push steam OUT, so it fries crisp instead of soaking up fat. — McGee; López-Alt
- Batter, last-minute & cold.** Whisk 150g plain + 150g rice flour, 9g baking powder, 6g salt; whisk in 380g ice-cold lager + 40g vodka to a thin batter. Leave it lumpy. Use now.

Cold + barely-mixed = low gluten = shatter, not bread; the CO2 escapes if it sits. Rice flour fries glassy; vodka suppresses gluten. — López-Alt
- Dredge & fry.** Dust each piece in the 80g flour, dip in batter, let it drip, lower gently into the oil. Fry 4–6 min to deep gold; don’t crowd.

The flour dredge gives the batter grip; crowding drops the oil temp and turns it greasy. — López-Alt
- Drain, salt, serve.** Onto a rack, salt at once, serve immediately.

Done: deep golden, lacy and crisp, the fish just flaking inside. **Drain on a rack (not a plate), salt, serve at once** — it softens as it steams under itself.

Timing & chips — the batter is the only live step; portion fish and blanch chips ahead. Par-fry the fish pale, then finish-crisp to order across a staggered crew dinner. Chips: blanch at 140°C, crisp at 185°C.

ONE BATTER, MORE THAN FISH
The same cold batter fries **onion rings, battered sausage, halloumi, courgette** — same dredge, same 185°C oil. Make a double batch and run a fried spread.

ELEVATION		
TIER 1 — NO EXTRA TIME Vodka splash — crisper, drier crust Crisp lager — beats a heavy ale Ice the bowl — keep the batter cold	TIER 2 — WORTH IT All-rice batter — gluten-free, extra-crisp Tempura-light — ice water, minimal mix Curry sauce / mushy peas	TIER 3 — SERVICE Scraps — fried batter bits, salted Tartare from scratch Triple-cooked chips

TROUBLESHOOT

Greasy	Oil too cool / batter sat → 185°C, mix to order
Not crisp	Too much gluten / no rice flour / warm batter → keep cold, use rice flour
Batter slides off	Fish wet or no dredge → dry the fish, flour it first
Pale & soft	Oil cool / pan crowded → fewer pieces, recover the heat

CHARTER PREP & SAFETY

Batter	Last-minute only — mix per service
Ahead	Portion fish; blanch chips; par-fry pale
Finish	Crisp to order, 185°C
Fryer safety	Fill the pot ≤ 1/3; never leave it; kill the heat in a swell